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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student :

Programme: B.E / B.Tech | Slot :

Course Code: CSA15 | Faculty : Dr.Arul Raj A.M

Course Name: Cloud Computing and Big Data Analytics | Date:

Duration: 1:30 Hr | Max Marks: 50 Marks

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, S and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

Given data

(10) (CO2) (BL3)

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

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CIA 1

Evaluation of cloud computing and XaaS Models

Cloud Computing provides Computing resources, applications through the internet.

Development in hardware and internet technologies have made cloud computing possible.

Development in computing hardware:

High performance processors, virtualization technology, Large storage devices, High speed networks and powerful data centres.

Development in internet software.

web applications, REST APIs, web services and Container technology.

(i) Existing content management Application

IaaS:

IaaS stands for Infrastructure as a service.

=) Provides virtual machines

=) Provides storage and networking

=) Users control the operating system.
=) Users install and manage their own software.

(ii) New customer-Facing Application - PaaS
PaaS stands for Platform as a service
=) Provides development tools.
=) Provides runtime environments.
=) Supports application deployment.
=) Cloud provider manages servers and operating systems.

(iii) Customer Relationship management SaaS.
SaaS stands for Software as a service.
=) Ready-made software
=) Accessible through a browser
=) No installation required.

Advantages of XaaS.
=) Reduced infrastructure costs.
=) Flexible resource usage
=) Easy scalability.
=) Faster deployment
=) Less maintenance.

2) Server Consolidation:

Given data

Number of physical servers = 15

Average CPU utilization = 25%

Power consumption per old server = 350 W

Power consumption per new host = 1450 W

Target CPU utilization = 65%

Electricity cost = $0.1/kWh

Operating hours = 24 × 365 = 8,760 hours

Fault tolerance = N+1.

(a) Number of active hosts and total hosts

Calculate total CPU workload.

$15 \times 25\% = 3.75$

Total workload = 3.75 server-equivalents.

$$\text{Hosts} = \frac{3.75}{0.65}$$

$$= 5.77$$

$$= 6$$

Including N+1 tolerance

6 + 1 = 7 hosts.

(b) Consolidation ratio

Consolidation ratio = $\frac{\text{old servers}}{\text{new hosts}}$

Based on six capacity - required active hosts:

$= \frac{15}{6}$

2.5 : 1

Including all seven installed hosts:

$\frac{15}{7} =$ **2.14 : 1**

(c) CPU utilization

when all seven hosts are active

$\frac{3.75}{6} \times 100$

$=$ **62.5%**

(d) Annual electricity cost saving

old power consumption.

$15 \times 350 = 5250\text{W}$

5.25KW

New power consumption

$7 \times 450 = 3150\text{W}$

3.15KW

old annual electricity cost

$5.25 \times 8760 \times 9$

₹4,13,910

New annual electricity cost

$3.5 \times 8760 \times 9 =$ **2,48,346**

Annual saving $= 413910 - 248346 =$ **1,65,564**

1,65,504

3. Virtualization - CPU and memory Analysis

Given

CPU cores = 24

RAM = 96 GB

Hypervisor RAM = 8 GB

VMs = 12

Each VM = 3 VCPUs and 8 GB RAM.

(a) Memory sufficiency

Total VCPUs :

$12 \times 3 = 36$

CPU ratio.

$36 / 24 = \boxed{1.5 : 1}$

memory ratio:

$96 / (96 - 8) = \boxed{1.09 : 1}$

(b) Memory sufficiency

Normal :

$12 \times 5 = 60$ GB

Available RAM = 88 GB

Free RAM = 28 GB. sufficient

peak:

$(8 + 8) + (4 \times 6) = 88$ GB

Free RAM = 0 GB.

exactly sufficient, with no reserve.

= 1,65,564

(c) CPU sufficiency

Normal:

$36 \times 0.35 = 12.6$

$12.6 / 24 \times 100 = 52.5\%$.

Peak:

$36 \times 0.75 = 27$

$27 / 24 \times 100 = 112.5\%$.

(d) Issues and Solutions

Issues	Solutions
CPU overload	Add CPU cores
memory shortage	Increase RAM
VM attacks	Hypervisor security
Data leakage	Encryption
Unauthorized access	MFA.

4. Cloud Federation between Two Universities.

Cloud Federation connects two universities to share research and digital library resources.

Conventional data Centre:

⇒ Physical servers

⇒ Storage

⇒ Networking

⇒ Virtual machine

SDDC:

⇒ Software - controlled infrastructure.

⇒ Automatic resource allocation

⇒ Better scalability

⇒ Centralized management.

Role of SOA:

⇒ Connects applications through APIs.

⇒ Shares research computing service.

⇒ Provides access to digital libraries.

Challenges:

1) Resource Sharing

2) Identity management

3) User Presence Privacy

4) Data Security.

Solutions:

⇒ multi-factor Authentication

⇒ Sigle sign-on.

⇒ Data encryption

⇒ Firewalls

⇒ Regular security audits

⇒ Data sharing agreements.

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5 Edge Computing and Cloud Storage:

Given

Cameras = 150

Clip size = 4.4MB

Interval = 3 minutes

Storage cost = $1.50/GB

serverless cost = $0.0005

Filtering = 25%.

(a) without Filtering

Clips per Camera:

$$\frac{1440}{3} = 480$$

Total clips:

$$480 \times 150 = 72,000$$

Daily data:

$$72,000 \times 4 = 288,000 \text{ MB}$$

$$= \boxed{288 \text{ GB}}$$

Bandwidth :

26.67 Mbps.

(b) monthly storage cost

288 × 30 × 1.50

= ₹ 12,960

(c) After filtering

Daily data

288 × 25% = 72GB

Bandwidth:

26.67 × 25% = 6.67 Mbps

monthly storage:

72 × 30 = 2160GB

storage cost:

2160 × 1.50 = ₹3,240

monthly savings:

12960 − 1.50 ₹3,240

(d) serverless cost

monthly invocations:

72,000 × 30 × 25% = 540,000

cost :

540,000 × 0.00015

= ₹81.

REST API:

Transfer data between Cameras and cloud storage using HTTP methods.

object storage:

Suitable for storing large video clips or video files, because it is scalable and cost effective.